

Capstone Defense — Complete Study Guide

Understand the whole project from the ground up: plain-English overview, every concept from scratch, the pipeline, the decisions, the numbers, and a large bank of likely panel questions with model answers.

PART A — The big picture

One sentence

I built a computer system that uses real rainfall and map data to estimate, for every small square of a flood-prone part of Kigali and for every day, whether flood pressure is Low, Moderate or High — and I checked those estimates against the government's official flood map.

30-second elevator pitch

Kigali floods repeatedly, but the official flood maps are static — they show generally risky zones, not how risk changes day by day after rain. I divided a flood-prone corridor into a 250-metre grid and, for every cell and every day from 2018 to 2024, combined real rainfall, terrain, rivers and buildings to classify flood pressure as Low, Moderate or High. Random Forest reached a macro-F1 of 0.83 on a future test year, beat simple baselines, and its high-pressure predictions fall 1.6 times more often inside the official flood zones than chance — even though it never saw those zones during training.

The 2-minute version

Start with the gap (static maps), state the idea (grid + daily rainfall + terrain → Low/Moderate/High pressure + an HMM for time), name the data (ERA5 rainfall, SRTM elevation, OpenStreetMap, official MOE polygons), give the headline numbers (RF 0.83 macro-F1, 0.87 high recall, baselines beaten, enrichment 1.59×), and finish with the contribution: a reproducible, interpretable, time-aware tool that reinforces and extends the official map, plus a dashboard. Be explicit that this is flood PRESSURE — a derived risk state — not confirmed flood events.

PART B — Every concept from scratch

Flood pressure vs flood event (know this cold)

A flood EVENT is observed water on the ground at a place and time. Flood PRESSURE is a modelled risk condition: how primed a place is to flood given recent rainfall, low/flat terrain, nearness to a river and exposure. I model pressure, not events, because dated, geolocated incident records were not openly available. This is stated in my scope and is why I validate against official risk polygons rather than incident reports.

Grid and cell-day

The study area is cut into 250 m squares (cells). A cell-day is one cell on one date — the basic row of my table. $729 \text{ cells} \times 2557 \text{ days} \approx 1.86 \text{ million cell-days}$.

Rainfall accumulation windows (1, 3, 7, 14 day)

Flooding depends not just on today's rain but on how much has fallen recently (the ground saturates). So for each day I sum rainfall over the past 1, 3, 7 and 14 days. These four numbers let the model see both sudden downpours and prolonged wet spells.

Terrain & exposure features — and why each matters

- Elevation (m): low-lying cells collect water — higher flood risk.
- Slope (deg): flat ground drains slowly and ponds; steep ground sheds water.
- Distance-to-river (m): near a watercourse = more exposed to overflow.
- Road & building density: proxies for urbanisation, sealed surfaces and drainage pressure / exposure.
- Official flood-polygon flag: whether the cell falls in the government's mapped flood zone.

Susceptibility (static) vs trigger (dynamic)

Susceptibility is the fixed, place-based part of risk (low, flat, near river, built-up). The trigger is the changing part (rainfall). Flood pressure = a heavy enough trigger acting on a susceptible enough place. My label multiplies a rainfall trigger by a susceptibility score, so heavy rain on a high, steep, well-drained hill still yields low pressure.

Supervised classification

Supervised learning means showing the model many examples with known answers (features → Low/Moderate/High) so it learns the pattern and can label new, unseen cell-days. It is classification because the answer is a category, not a number.

The models, in plain English

- Logistic Regression: a simple weighted-sum classifier; interpretable baseline.
- Decision Tree: a flowchart of yes/no questions on the features; interpretable but can overfit.
- Random Forest: hundreds of decision trees voting; robust and strong on tabular data — my best model.
- XGBoost: trees built in sequence, each fixing the previous errors (gradient boosting); top-tier on tabular data.

Hidden Markov Model (HMM)

An HMM models a sequence that moves between hidden states over time. Three ideas: (1) hidden STATES (here, latent flood-pressure regimes); (2) OBSERVATIONS emitted by each state

(the rainfall features); (3) a TRANSITION MATRIX giving the probability of moving from one state to another the next day. Fitting (Baum-Welch algorithm) learns these from data; decoding (Viterbi) infers the most likely state sequence. My transition matrix is strongly diagonal ($\approx 0.83/0.80/0.91$), which is the model saying flood pressure persists once it sets in. It adds the time dimension the per-day classifier lacks.

Evaluation metrics

- Accuracy: fraction of cell-days classified correctly.
- Precision (of a class): of those predicted High, how many really were High.
- Recall (of a class): of those that really were High, how many we caught. High recall matters most — missing a flood is worse than a false alarm.
- F1: the balance (harmonic mean) of precision and recall.
- Macro-F1: the average F1 across all three classes, weighting each class equally — fair when classes are imbalanced.
- Confusion matrix: a table of true vs predicted classes showing exactly where errors happen.

Temporal hold-out (why not a random split)

A random split can let the model peek at days near the test days and inflate scores. I instead train on 2018–2023 and test on the entire unseen 2024. Doing well on a future year is real evidence the model generalises, not memorises.

SHAP and feature importance

These tell us which inputs drive predictions. SHAP fairly attributes each prediction to its features. In my model, 3- and 7-day rainfall dominate, then low elevation and nearness to a river — exactly the physics of flooding, which makes the model trustworthy and auditable.

Baselines

A baseline is a deliberately simple method you must beat to prove your complex model earns its place. Mine are a rainfall-only threshold and a static-polygon flag. Beating both (0.83 vs 0.62 and 0.32) shows the geospatial context and learning add genuine value.

Spatial validation: containment, enrichment, odds ratio

Containment = share of predicted-High cells inside official polygons (0.30). Because the official zone covers only ~19% of my corridor, a high containment is geometrically impossible without cheating, so I report enrichment = containment $\div$ base rate (1.59 $\times$, i.e. 59% more concentrated than chance) and the inside-vs-outside odds ratio (1.84 $\times$). These measure agreement fairly given a sparse official layer.

The data products

- ERA5 (via Open-Meteo): a reanalysis — observations blended into a physics model to give continuous global weather, including daily rainfall. Keyless and reproducible.
- CHIRPS: satellite + station rainfall at 0.05° (~5.5 km), tuned for Africa; scientifically ideal but harder to access here.
- SRTM: Shuttle Radar Topography Mission — global elevation, the source for elevation and computed slope.
- OpenStreetMap: crowd-sourced map data for rivers, roads and buildings, queried through the Overpass API.
- Official polygons: Rwanda GeoPortal / geodata.rw, Ministry of Environment flood-risk areas.

PART C — The pipeline, step by step

Everything runs from main.py: build → train → evaluate → dashboard. Stages cache their data, so re-runs are fast.

build_real_dataset.py

1. Define the corridor and build the 250 m grid (729 cell centroids).
2. Fetch SRTM elevation per cell; compute slope from the elevation grid.
3. Query OpenStreetMap for waterways, roads and buildings; compute distance-to-river and road/building densities per cell.
4. Compute a static susceptibility surface; set the official-polygon flag from the real MOE layer (point-in-polygon).
5. Fetch ERA5 daily rainfall (2018–2024); compute 1/3/7/14-day sums.
6. Assemble the cell-day table and derive Low/Moderate/High labels.

Label construction & the non-circularity trick (critical)

Label score = rainfall trigger × (terrain susceptibility) + an unobserved drainage latent + daily noise. The latent and noise are NOT given to the model. Without them the label is a closed formula the model could perfectly reverse-engineer (an early run scored 0.99 F1 — a red flag). With them, performance is a realistic 0.83. High is the top 20% of scores so it is a genuine severe tail, not a third of all days.

train_real.py

Splits 2018–2023 (train) vs 2024 (test); trains the baselines, Logistic Regression, Decision Tree, Random Forest and XGBoost; fits the HMM on per-cell rainfall sequences; saves models, confusion matrices, SHAP, the transition matrix and results_summary.json.

evaluate_spatial.py

Loads the trained XGBoost, predicts 2024, and computes the spatial validation (containment, enrichment, odds ratio) plus the per-cell predicted-High-frequency risk map.

[dashboard.py](#)

A Streamlit app: pick a date, see the predicted map, overlay official polygons, inspect any cell's features, probabilities and rainfall history, and view model/HMM performance panels.

PART D — Key decisions and how to justify them

Q. Why this corridor and 250 m grid?

A. It is a known flood-prone Kigali corridor; I shifted it south to overlap the real official flood zone. 250 m is fine enough to resolve local variation and coarse enough to stay computationally feasible (the proposal allowed 100–250 m).

Q. Why 2018–2024?

A. A long, continuous daily rainfall record gives multiple wet/dry seasons and lets me hold out a full future year (2024) for honest testing.

Q. Why ERA5 and not CHIRPS or Meteo Rwanda?

A. ERA5 via Open-Meteo is the only source I can fetch reproducibly and keylessly; I demonstrated it live. CHIRPS is scientifically preferable but I tested two access routes (IRI Data Library, ClimateSERV) and both failed without heavy tooling. Meteo Rwanda is authoritative but has no open API. So ERA5 now, CHIRPS/Meteo Rwanda as documented future work.

Q. Why derived labels, and why add noise?

A. Open incident data was unavailable, so I derive flood pressure from rainfall and terrain — as my scope permits. I add an unobserved drainage/noise term so the task is realistic; otherwise the model just inverts a formula and scores an artificially perfect 0.99.

Q. Why Random Forest and XGBoost?

A. They are the strongest, most robust models for tabular geodata and they give feature importance/SHAP for interpretability.

Q. Why an HMM on top?

A. The classifier judges each day independently; the HMM adds the time dimension — how pressure persists and transitions — which is central to a day-to-day warning tool.

Q. Why reframe the 70% spatial target?

A. The official national layer covers only ~19% of the corridor, so 70% containment is mathematically impossible unless I make the label depend on the polygon, which would be circular. Enrichment and odds ratio measure the same intent honestly.

PART E — Numbers to memorise

- Grid: 729 cells, 250 m, corridor in Kigali. Period: 2018–2024 daily = 2557 days.
- Rows: 1,864,053 cell-days. Test year 2024 = 266,814 rows.
- Official flood polygons: 136 / 729 cells $\approx$ 19%.
- Random Forest: macro-F1 0.831, accuracy 0.847, high-recall 0.867.
- XGBoost: 0.813 / 0.847. Decision Tree 0.753. Logistic Regression 0.749.
- Baselines: rainfall-threshold 0.623, static-polygon 0.324.
- HMM self-transitions $\approx$ 0.83 / 0.80 / 0.91; next-step accuracy 0.45 (random 0.33).
- Spatial: containment 0.30, enrichment 1.59 $\times$, inside/outside odds 1.84 $\times$ (30.7% vs 16.7%).
- Live ERA5 check: 131.2 mm rainfall, AOI centroid, January 2024.

PART F — Anticipated panel questions (with model answers)

DATA

Q. How do I know your data is real, not made up?

A. Every source is fetched live from a public API and cached: ERA5 rainfall, SRTM elevation, OpenStreetMap, and the government flood polygons from geodata.rw. I can run the fetch in front of you — for example a live ERA5 pull gives 131.2 mm for my area in January 2024 with the real daily values.

Q. Isn't ERA5 too coarse for a small corridor?

A. Yes, ERA5 is $\sim$ 9–25 km, so rainfall varies more by date than by cell across a few km. That is acceptable because the spatial variation in flood pressure comes from terrain and hydrology, which I have at fine resolution; rainfall supplies the temporal trigger. Finer CHIRPS/gauge data is future work.

Q. Why are road/building densities relevant to flooding?

A. They proxy urbanisation: sealed surfaces increase runoff and reduce infiltration, and dense settlement raises exposure and drainage pressure.

METHODOLOGY

Q. Is your label circular — features predicting a label made from features?

A. I anticipated this. The label is derived from rainfall and terrain, but I add an unobserved drainage latent and daily noise the model never sees, so it cannot perfectly invert the formula. The proof is that performance is a realistic 0.83, not the 0.99 an early circular version produced.

Q. Why three classes and why is High only 20%?

A. Low/Moderate/High matches how warnings are communicated, and concentrating High in the top 20% makes it a genuine severe tail rather than a third of all days — closer to reality and a stricter target for the model.

Q. Could you predict actual flood events instead of pressure?

A. Only with dated, geolocated incident records, which were not openly available. That is the headline item in my future work, ideally via MINEMA records.

MODELS & EVALUATION

Q. Why does Random Forest beat XGBoost here?

A. They are very close (0.831 vs 0.813). On this tabular, moderately noisy problem Random Forest's averaging is slightly more robust; with tuning XGBoost often matches or exceeds it — hyper-parameter search is future work.

Q. Why macro-F1 and high-recall rather than accuracy?

A. Classes are imbalanced and missing a high-pressure cell is the costly error, so I weight each class equally (macro-F1) and report recall on the High class specifically.

Q. How do you know it generalises?

A. I test on a completely unseen future year (2024) after training on 2018–2023, not a random split, so leakage between adjacent days cannot inflate the score.

Q. Your spatial agreement is only 0.30 — isn't that a failure?

A. No — it is the honest, expected result and a finding in itself. The official national map covers only 19% of the corridor, so raw containment is capped. The fair measures are enrichment (1.59×) and odds ratio (1.84×): the model concentrates High inside official zones far more than chance, and extends them with a dynamic footprint — which is exactly the gap the project addresses.

LIMITATIONS & IMPACT

Q. What are the main limitations?

A. Derived (not event-grounded) labels; ERA5 coarser than gauges; and a low-resolution official polygon layer. None invalidate the framework but each bounds the claims, and each has a concrete future-work response.

Q. Who would actually use this and how?

A. City and disaster officers, to prioritise inspection, drainage clearing and community alerts in the days after heavy rain — through the dashboard, by date and location. It supports, not replaces, official early warning.

Q. What would you do with more time or data?

A. MINEMA incident validation, CHIRPS/Meteo Rwanda rainfall, TWI and HAND hydrological features, Optuna tuning, and an operational daily-ingestion dashboard.

TECHNICAL DEEP-DIVE

Q. Explain the HMM transition matrix in your own words.

A. Each row is the current state, each column the next-day state, and the values are probabilities that sum to one. Mine is strongly diagonal — about 0.83, 0.80 and 0.91 on the diagonal — meaning a cell most likely stays in its current pressure state the next day, capturing real persistence after rain.

Q. How did you compute distance-to-river without GIS software?

A. I densified the OpenStreetMap river geometries into points, projected to metres, and used a KD-tree nearest-neighbour query from each cell centroid — pure NumPy/SciPy, no heavyweight GIS dependency.

Q. How is the dashboard wired to the model?

A. It loads the saved XGBoost model and the dataset, predicts for the selected date, and renders the grid with pydeck, overlaying the official polygons and exposing per-cell probabilities and rainfall series.

CURVEBALLS

Q. If a panellist finds a genuine flaw you missed —

A. Acknowledge it directly, say why it does not overturn the core result if true, and frame the fix as future work. Honesty and composure score better than defensiveness.

Q. If you are asked something you do not know —

A. Say so plainly, then reason aloud toward a sensible answer: ‘I haven’t tested that, but based on X I would expect Y, and I would verify it by Z.’

PART G — Handling tough moments

- Pause, breathe, restate the question in your own words before answering.
- Lead with the one-line answer, then justify — do not narrate your way to it.

- If challenged, agree where the point is fair, then defend the core with a number.
- Never bluff. 'I don't know' plus reasoning beats a confident wrong answer.
- Bring every answer back to the contribution: time-aware pressure that reinforces and extends the static map.

PART H — Live demo checklist

7. Open a terminal in the project folder; activate the virtual environment.
8. Show a live data pull (ERA5) to prove the data is real.
9. Run: `python main.py dashboard`.
10. Pick a wet date, show the predicted map, toggle the official-polygon overlay.
11. Click a high-pressure cell; show its rainfall history and probabilities.
12. Open `results_summary.json` / the figures if asked for evidence.

Have the slides PDF and a screen-recording of the dashboard as backups in case the live demo fails.

PART I — Glossary

AOI: Area of Interest — the study corridor.

Cell-day: One grid cell on one date; the basic data row.

Reanalysis (ERA5): Observations blended into a physics model to give continuous weather.

SRTM: Satellite mission providing global elevation data.

Overpass: The query API used to pull OpenStreetMap data.

Susceptibility: Static, place-based flood-proneness (low, flat, near river, built-up).

Trigger: The dynamic part of risk — rainfall.

Macro-F1: Average F1 across classes, weighting each class equally.

Recall: Of the true positives, the fraction the model caught.

Temporal hold-out: Testing on a future period unseen during training.

HMM: Hidden Markov Model — models hidden states evolving over time.

Transition matrix: Probabilities of moving between states from one step to the next.

SHAP: A method that attributes a prediction to its input features.

Enrichment / lift: How much more concentrated predicted-High is in official zones than chance.

Containment: Share of predicted-High cells inside official polygons.